Javascript API

AIM:

To fetch and display data from a web API using JavaScript.

DESCRIPTION:

This program demonstrates how to fetch data from an external API using JavaScript and display it on a web page. The webpage contains a button that, when clicked, retrieves user data from a public API using the fetch() method.

The fetched data is in JSON format and is converted into a JavaScript object. Specific details such as name, email, and city are extracted and displayed dynamically inside the webpage. CSS is used to style the layout and make it visually appealing.

Error handling is also included using the catch() method to display a message if the API request fails. This program helps in understanding how JavaScript interacts with APIs to build dynamic and data-driven web applications.

ALGORITHM:

1. Start the program.
2. Create the basic HTML structure using <html>, <head>, and <body>.
3. Add a title for the webpage.
4. Apply CSS styling to design the layout and appearance.
5. Create a container to hold the content.
6. Add a heading to describe the API example.
7. Add a button to trigger data fetching.
8. Create a element to display the fetched data.
9. Define a JavaScript function getData() to fetch data from the API.
10. Use the fetch() method to request data from the given API URL.
11. Convert the response into JSON format using .json() method.
12. Extract required fields such as name, email, and city from the data.
13. Display the extracted data inside the result using innerHTML.

14. Use .catch() method to handle errors during the fetch process.
15. Display an error message if data fetching fails.
16. Execute the function when the button is clicked.
17. End the program.

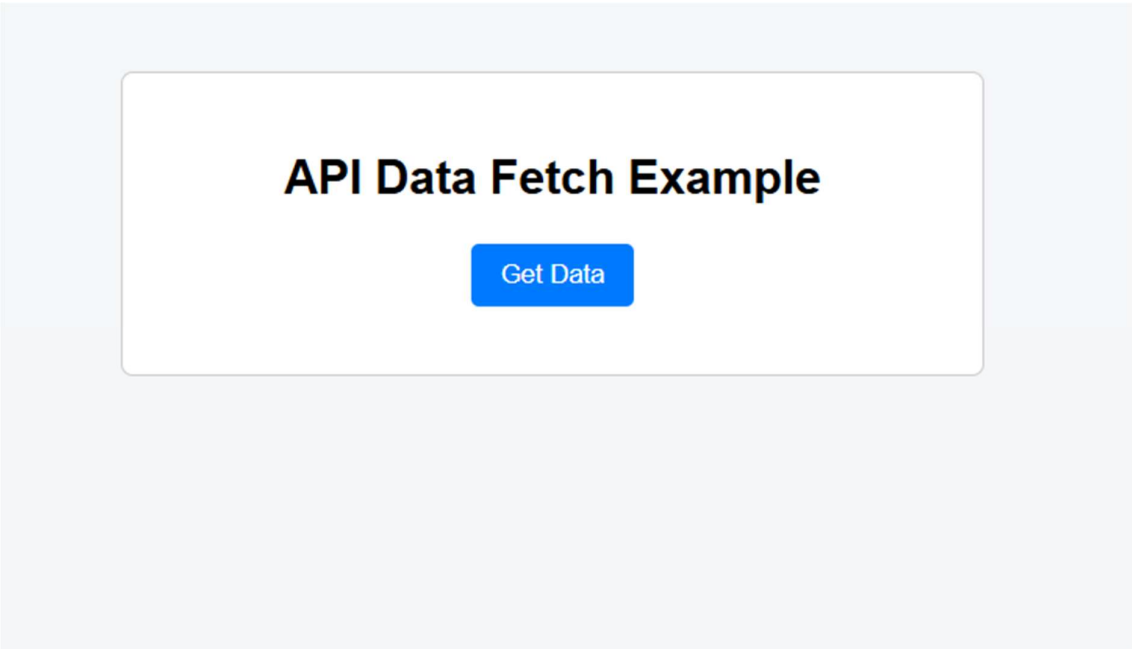
CODE:

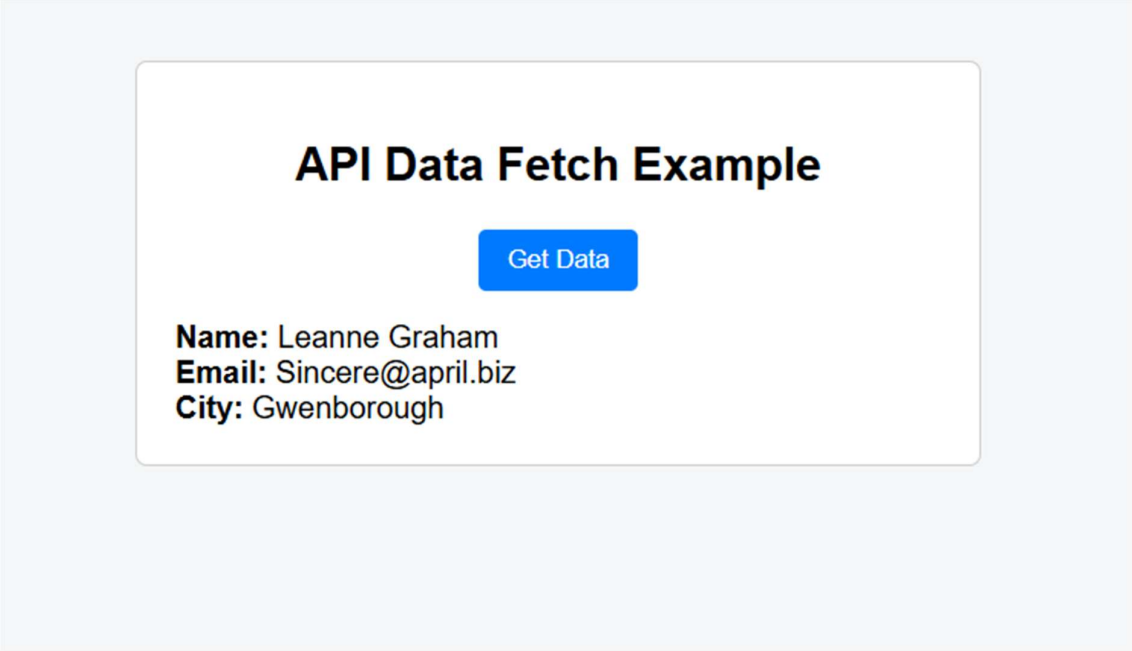
```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript API Example</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f6f8;
      text-align: center;
      margin-top: 60px;
    }

    .container {
      width: 400px;
      margin: auto;
      padding: 20px;
      background: #ffffff;
      border: 1px solid #ccc;
      border-radius: 6px;
    }
```

```
      button {
        padding: 8px 15px;
        background-color: #007BFF;
        color: white;
        border: none;
        border-radius: 4px;
        cursor: pointer;
      }

      button:hover {
        background-color: #0056b3;
      }

      .data {
        margin-top: 15px;
        text-align: left;
      }
    </style>
  </head>

  <body>

  <div class="container">
    <h2>API Data Fetch Example</h2>
    <button onclick="getData()">Get Data</button>

    <div id="result" class="data"></div>
  </div>
```

```
<script>
function getData() {
  fetch("https://jsonplaceholder.typicode.com/users/1")
  .then(response => response.json())
  .then(data => {
    document.getElementById("result").innerHTML =
      "<b>Name:</b> " + data.name + "<br>" +
      "<b>Email:</b> " + data.email + "<br>" +
      "<b>City:</b> " + data.address.city;
  })
  .catch(error => {
    document.getElementById("result").innerHTML = "Error fetching data";
  });
}
</script>

</body>
</html>
```

OUTPUT:


API Data Fetch Example
Get Data


API Data Fetch Example
Get Data
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